SITHUM MANUSHA

SOFTWARE ENGINEERING INTERN

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ABOUT ME

Third-year undergraduate studying Information Technology and Management at the University of Moratuwa, passionate about software development and building reliable web applications. Hands-on experience developing full-stack projects using React, Next.js, Java Spring Boot, and modern PHP (MVC). Seeking a Software Engineering Internship where I can apply my problem-solving skills, learn from senior engineers, and contribute to real-world software products.

EDUCATION

University of Moratuwa (UOM) - Moratuwa, Sri Lanka **2024 - Present**
B.Sc. (Hons) in Information Technology & Management **(3rd Year | CGPA: 3.23 / 4.0)**

Open University of Sri Lanka (OUSL) - Nawala, Sri Lanka **2025 - Present**
Bachelor of Laws Honours Degree Programme (LL.B.)

St. Joseph's Vaz College - Wennappuwa, Sri Lanka **2009 - 2022**
G.C.E. Advanced Level (2 A's & 1 B) | G.C.E. Ordinary Level (8 A's & 1 B)

FEATURED PROJECTS

1. Super POS — Multi-Branch Enterprise Offline-First Retail System (Group Project) **2025 – 2026**

- Architected an offline-first transaction queue allowing branch cashiers to continue sales during network disruptions, buffering records with auto-sync in client-side **IndexedDB**.
- Implemented automated background synchronization featuring **Last-Write-Wins (LWW)** conflict detection and an interactive three-way resolver for cross-branch consistency.
- Engineered a real-time node fleet monitor with **Socket.io** telemetry for regional status tracking, fleet-wide broadcast alerts, and automated audit report exports.
- **Technologies:** Next.js, React, TypeScript, NestJS, Node.js, PostgreSQL, Prisma ORM, Socket.io, IndexedDB, Tailwind CSS, Turborepo
- **Live Demo:** pos-system-pos-web-app.vercel.app | **GitHub:** github.com/Ryzera/Ryzera-POS

2. CareerFlow — Full-Stack ATS Resume Optimizer & AI Interview Platform (Individual Project) **2025 – 2026**

- Engineered a dual-scoring ATS resume audit engine utilizing **Scikit-learn TF-IDF vectorization & Cosine Similarity**, computing 0–100% job match calibrations and automated bullet rewrites.
- Developed an interactive 5-round technical mock interview simulator leveraging **Google Gemini 1.5 Flash GenAI** to evaluate candidate answers against STAR rubrics.
- Constructed a decoupled microservices architecture featuring a **Java 21 Spring Boot 3** gateway with token-bucket rate limiting (10 req/min), multi-stage **Docker CI/CD**, and 26 automated tests.
- **Technologies:** Java 21, Spring Boot 3, Python 3.12, FastAPI, Next.js 14, Google Gemini AI, PostgreSQL, Docker, GitHub Actions, JUnit 5, Pytest
- **Live Demo:** careerflow-pink.vercel.app | **GitHub:** github.com/SithumManusha/careerflow

3. Ledgerly — Autonomous AI Financial Intelligence & Collaborative Ledger (Individual Project) **2025 – 2026**

- Engineered an AI financial tracking copilot evaluating composite 0–100 financial health scores, detecting spending velocity anomalies, and delivering actionable financial insights.
- Developed an interactive runway simulator using **Recharts** to stress-test discretionary spending cuts and capital purchases against comparative 6-month cash-flow curves.
- Solved multi-party debt disputes via a **Greedy Two-Pointer debt-minimization algorithm** across 4 split models (Equal, Percentage, Exact, Occupancy Days) with real-time SSE alerts and streaming PDF settlements.
- Fortified with end-to-end type safety, OWASP session security, and 68 automated unit/integration tests (**Vitest**).
- **Technologies:** React, TypeScript, Node.js, Express, PostgreSQL, Drizzle ORM, SSE, Recharts, Vitest, Render
- **Live Demo:** ledgerly-nine-rust.vercel.app | **GitHub:** github.com/SithumManusha/Ledgerly

4. Pulse AI Blog Engine — AI Content Platform & MVC Architecture (Individual Project) **2025 – 2026**

- Engineered an AI-native content publishing platform featuring an integrated intelligent assistant for one-click executive summaries, article drafting, and automated taxonomy tagging.
- Integrated an in-browser audio narration player with variable speed controls to deliver accessibility-focused, hands-free article listening via **Web Speech API**.
- Hardened application security using **PDO prepared statements** against SQL injection, session-based form protection, HTML sanitization, and automated **CodeQL** vulnerability scanning.
- **Technologies:** PHP 8, MySQL, SQLite (PDO), Google Gemini AI, Docker, GitHub Actions, CodeQL, Tailwind CSS, Web Speech API
- **Live Demo:** pulse-ai-blog-engine.onrender.com | **GitHub:** github.com/SithumManusha/pulse-ai-blog-engine

5. Automated Robotic Sorting System — Microcontroller Conveyor Platform (Group Project) **2024 – 2025**

- Constructed an automated conveyor sorting platform with an articulated 4-DOF robotic arm and sensory array to detect, categorize, and sort physical items into target collection bins.
- Programmed motor calibration and sensory feedback routines to achieve 98% placement precision across diverse tagged object categories.
- Synchronized mechanical arm pick-and-place actuation with continuous conveyor belt movement through hardware timing and integration testing.
- **Technologies:** Arduino Mega/Uno (C++), Stepper Motors (NEMA 17), Servo Motors, RFID, Sensors, Integration Testing
- **GitHub:** github.com/SithumManusha/robotic-sorting-system

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript, TypeScript, PHP, C/C++, SQL
- **Front-end Development:** React, Next.js, TypeScript, Tailwind CSS, Recharts, IndexedDB
- **Back-end Development:** Spring Boot 3, FastAPI, Node.js, Express, NestJS, PHP (MVC), WebSockets (Socket.io), RESTful APIs
- **Databases & ORM:** PostgreSQL (Supabase), MySQL, SQLite, Spring Data JPA, Hibernate, Prisma ORM, Drizzle ORM, PDO
- **DevOps, Testing & Quality:** Docker, Docker Compose, Git, GitHub Actions (CI/CD), CodeQL, JUnit 5, Pytest, Vitest, Playwright, Postman

CORE COMPETENCIES & ARCHITECTURE

MVC & Microservices Architecture, RESTful API Design, Agile / Scrum, Object-Oriented Design (SOLID), Clean Code, Automated Testing, Offline-First Sync

CERTIFICATIONS

- AWS Cloud Practitioner Essentials – Amazon Web Services (AWS)
- Software Testing with Generative AI – Simplilearn
- Introduction to Artificial Intelligence – Simplilearn
- Introduction to Cybersecurity – Cisco Networking Academy

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Member of the Organizing Committee, Official Career Fair conducted by University of Moratuwa (2026)
- Member of the Organizing Committee, Project "Rest with Hope" conducted by Leo Club, UoM (2025)
- Active Member – FIT Moments (Faculty Media Body)
- Active Member – Leo Club of University of Moratuwa
- Active Member – Legal Youth Community Sri Lanka

LANGUAGES

- English: Professional Working Proficiency (Oral & Written) | Sinhala: Native Proficiency

REFERENCES

Dr. A.L.A.R.R. Thanuja (B.Sc. (Sri Lanka), Ph.D. (USA)),
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